Micro-Seismic Monitoring and Roof Fall Risk Prediction using Acoustic Emission Telemetry in Underground Coal Mines

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1. Abstract

Roof fall accidents in underground bord-and-pillar coal mines cause critical casualties.
This paper presents a multi-channel acoustic emission telemetry network for real-time strata movement monitoring.
A total of 1.8 million micro-seismic events were recorded over a 12-month monitoring campaign..

2. Methodology & Machine Learning

We applied Convolutional Neural Networks (CNN) and Support Vector Regression (SVR) to classify acoustic emission energy c
The sensor threshold was calibrated to an acoustic emission amplitude of 45 dB..

3. Results

The micro-seismic roof fall forecasting model provided an early warning lead time of 36 hours prior to main roof collapse.
The model achieved an overall accuracy of 92.5%, precision of 90.4%, recall of 92.6%, and F1-score of 0.915..